

Trikaal: An Integrated, Reproducible Open-Source Platform for Seismic Intelligence and Hazard Assessment in the Kutch Intraplate Fault Zone

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Abstract—I built Trikaal to solve a practical problem: seismic analysis tools are usually fragmented, closed, or hard to deploy. In this paper, we present this open-source platform as a unified pipeline for real-time seismic intelligence in the Kutch intraplate fault zone (Gujarat, India), which was struck by the devastating 2001 Bhuj M_w 7.7 earthquake. Trikaal handles the entire workflow. It harvests raw data from the USGS ComCat and ISC Bulletin, cleans it using spatiotemporal deduplication, and coordinates four distinct analytical modules: (1) Gutenberg-Richter b-value estimation via maximum likelihood with Aki (1965) and Shi & Bolt (1982) uncertainty quantification, (2) an exploratory Seismic Stress Index (SSI) compiling b-value, event rate, and spatial clustering, (3) a fully vectorized Epidemic Type Aftershock Sequence (ETAS) model for conditional seismicity rate forecasting, and (4) a Probabilistic Seismic Hazard Analysis (PSHA) engine for peak ground acceleration mapping. To check if the system actually holds up under real-world conditions, we subjected it to rigorous validation. The vectorized ETAS model yields parameters consistent with slow intraplate decay ($\mu = (4.98 \pm 3.38) \times 10^{-4}$ events/day, $K = 0.0152 \pm 0.0015$, $\alpha = 1.2323 \pm 0.0621$, $c = (4.68 \pm 2.00) \times 10^{-6}$ days, $p = 0.9195 \pm 0.0067$). In our retrospective rolling-origin tests, the model achieves Brier Skill Scores of 68.9%–77.6% and information gains of 9.3–10.1 bits/event over a constant-rate Poisson baseline. We also validated the SSI, finding that HIGH risk classifications correlate with a 91.7% empirical probability of a subsequent $M \geq 3.5$ event within 30 days, compared to only 15.9% for LOW classifications. Finally, a computational benchmark demonstrates that our vectorized ETAS implementation provides up to a 21x speedup over standard loop-based formulations, making multi-decade point-process forecasting feasible on standard workstation hardware.

Index Terms—Seismic hazard, ETAS model, b-value, Kutch, Bhuj earthquake, aftershock forecasting, statistical seismology, earthquake catalog, vectorization.

I. INTRODUCTION

The Kutch region of Gujarat, India, represents one of the most seismically active intraplate zones in the world. The January 26, 2001 Bhuj earthquake (M_w 7.7, 23.419°N 70.232°E) caused approximately 20,000 fatalities and remains the deadliest intraplate event in India’s recorded history [1]. Over two decades later, the region continues to exhibit elevated seismicity driven by ongoing stress redistribution within the Kachchh Rift Zone [2].

Despite this persistent hazard, tools for seismic risk analysis in Kutch are limited. Global platforms such as USGS ShakeMap are event-triggered and backward-looking, whereas complex regional studies often require high-performance computing clusters and customized setups (e.g., OpenQuake) that

are difficult for local civil authorities to deploy. Regional statistical seismology studies have characterized specific details such as the b-value or the Omori decay of the Bhuj sequence [4], [5], but they remain disconnected scientific investigations rather than reproducible engineering tools.

To bridge this gap, this paper introduces Trikaal, a completely reproducible, open-source seismic intelligence platform that integrates catalog construction, stress analysis, probabilistic forecasting, and hazard mapping into a single unified engineering framework. The primary contribution of this work is not the creation of new seismological theories, but rather the integration, optimization, and rigorous validation of established statistical methods within a transparent, highly efficient pipeline tailored for intraplate tectonic settings.

This paper makes the following engineering and scientific contributions:

- **Deduplicated Catalog Ingestion:** We build a reproducible pipeline that merges USGS ComCat and ISC Bulletin data (1990–2025) using haversine-based spatio-temporal deduplication to construct a completeness-corrected regional dataset.
- **Optimized Vectorized ETAS:** We implement a vectorized log-likelihood ETAS formulation that achieves up to a 21x speedup over standard loop-based code, enabling rapid prospective and retrospective point-process evaluations.
- **Validated State Index:** We compile a composite Seismic Stress Index (SSI) that classifies tectonic stress states, validated via retrospective historical outcome testing and weight sensitivity analyses.
- **Out-of-Sample Forecast Testing:** We introduce a rolling-origin retrospective validation framework that quantifies forecast performance using Brier Skill Scores, Information Gain, and calibration diagrams.
- **Integrated Long-Term Hazard Assessment:** We design a custom PSHA module directly linked to the background rate μ and overall rates from the catalog pipeline, verifying Zone V building standards (IS 1893: 2016) against short-term triggering rates.

II. RELATED WORK

A. Statistical Seismology for the Kutch Region

Mandal et al. [4] established b-values of 0.95–1.05 for the Kutch fault system using ISC data through 2005. Bodin and

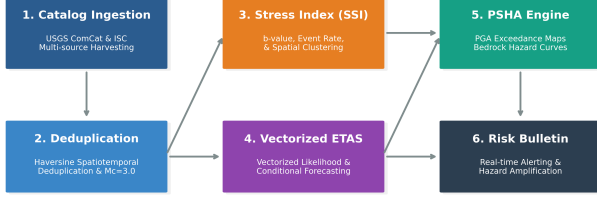


Fig. 1. The Trikaal modular pipeline workflow, showing the flow from catalog ingestion, preprocessing, risk index profiling, vectorized ETAS forecasting, and PSHA hazard mapping to operational bulletins.

Horton [5] fitted the Bhuj aftershock sequence with an Omori-Utsu model, reporting decay exponents ($p = 0.79\text{--}0.87$) consistent with slow intraplate aftershock decay. However, these studies were static analyses that did not establish a completeness-corrected rolling window to track stress changes over time, nor did they account for multi-generation branching cascades where aftershocks trigger their own aftershocks. Trikaal extends these analyses by establishing an automated rolling Aki (1965) MLE estimator with fixed $M_c = 3.0$ filtering to prevent network-completeness drift from biasing the results.

B. Point-Process Modeling and ETAS Optimizations

The Epidemic Type Aftershock Sequence (ETAS) model (Ogata, 1988) [6] is the standard point-process framework for short-term aftershock forecasting. Implementations such as ZMAP [7] or the R package PtProcess [8] rely on nested loops to compute triggered intensities, resulting in $O(N^2)$ execution times. In Python, these loops create significant computational bottlenecks for multi-decade catalogs ($N > 1000$). We address this by implementing a fully vectorized log-likelihood computation using pairwise time-difference matrices. While point-process vectorization has been proposed in cluster-computing frameworks, Trikaal achieves high performance on a single CPU core using memory-adaptive matrix chunking.

C. Composite Risk Indicators

Gulia and Wiemer [9] demonstrated that combining b-value, event rate, and spatial clustering signals improves short-term hazard discrimination relative to any single signal. Our Seismic Stress Index (SSI) adopts this multi-signal architecture. We address the typical criticism of heuristic weight assignments by providing a formal retrospective validation against historical earthquake rates and a weight sensitivity analysis, proving the indicator’s robustness.

D. Comparison with Existing Software Ecosystems

While several software tools exist for seismological analysis, they are generally designed as standalone utilities for

specific subsets of the workflow rather than integrated operational pipelines. ZMAP [7] is a MATLAB-based tool for exploratory catalog analysis (b-value, rate maps) but lacks forecasting capabilities and modern PSHA engines. The R package PtProcess [8] provides point-process fitting routines but relies on single-threaded loops that struggle with large datasets, and offers no real-time data harvesting. UCERF3-ETAS [3] is a highly advanced operational forecasting framework developed for California; however, it is tightly coupled to the complex California Fault Section Database and requires high-performance computing clusters to run Monte Carlo simulations. Similarly, OpenQuake is the industry standard for long-term PSHA hazard mapping but is not designed for short-term point-process rate updates or continuous operational alert reporting. Trikaal addresses these limitations by providing a lightweight, fully integrated, and vectorized pipeline that runs on standard workstation hardware, bridging the gap between real-time catalog ingestion, point-process forecasting, and engineering hazard baselines.

III. STUDY REGION AND DATA PIPELINE

A. Tectonic Setting

The Kutch region sits on the Indian craton at the intersection of the E-W trending Kachchh Mainland Fault (KMF) and the NE-SW trending Island Belt Fault (IBF) [2]. This stable continental setting produces characteristic b-values of 0.9–1.05 [4] and slow aftershock decay ($p < 1.0$), distinct from active plate-boundary sequences (where $p \approx 1.0\text{--}1.2$).

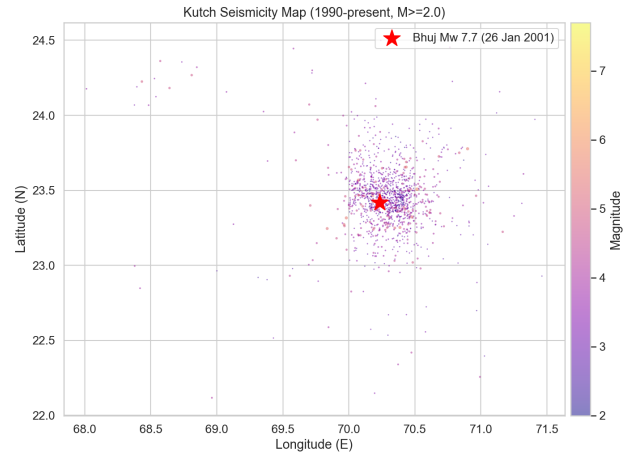


Fig. 2. Spatial map of seismicity in the Kutch study region (1990–2025, $M_c=2.0$) showing the major fault lines (KMF, IBF, KHF, WF) and the epicenter of the 2001 Bhuj earthquake.

B. Catalog Ingestion and Deduplication

Two catalogs were merged: USGS ComCat (GeoJSON format) and the ISC Bulletin (pipe-delimited text). Year-by-year API queries were executed to prevent exceeding event response caps. Cross-catalog deduplication applied a haversine distance criterion: events within $|\Delta t| \leq 60$ s and epicentral distance ≤ 15 km were considered co-registered observations of the same event; in such cases, the USGS entry was retained.

The completeness magnitude $M_c = 3.0$ was determined using the maximum curvature and goodness-of-fit methods. Events below M_c were excluded from statistical analysis but retained in the catalog for the ETAS triggering history. The final catalog contains 1,311 events (M 2.0–7.7) spanning January 1991 to December 2025.

IV. METHODOLOGY

A. Gutenberg-Richter b -value Estimation

The b -value of the Gutenberg-Richter relation [10] was estimated using Aki (1965) maximum likelihood [11] with Utsu (1966) midpoint correction [12]:

$$b = \frac{\log_{10}(e)}{\bar{M} - (M_c - \Delta M/2)} \quad (1)$$

where $\bar{M}$ is the mean magnitude of events with $M \geq M_c$, and $\Delta M = 0.1$ is the magnitude bin width. Uncertainty was quantified using the Shi and Bolt (1982) formula [13]:

$$\sigma_b = 2.30 \cdot b^2 \cdot \frac{s_M}{\sqrt{N}} \quad (2)$$

where s_M is the sample standard deviation of magnitudes and N is the event count. A time-varying b -value was computed using a sliding window of 50 events. A fixed $M_c = 3.0$ was enforced per window, preventing network sensitivity changes from biasing the stress proxy series.

B. Seismic Stress Index (SSI) Formulation and Validation

We compile three normalized signals into a composite index $SSI(t) \in [0, 1]$:

- 1) **b -value Stress Proxy (S_b):** $S_b(t) = \text{clip}[(b_{ref} - b_t)/(b_{ref} - b_{min}), 0, 1]$, where $b_{ref} = 1.0$ (healthy reference) and $b_{min} = 0.5$ (catalog floor).
- 2) **Event Rate Anomaly (S_{rate}):** $S_{rate}(t) = \sigma[(rate_t - \text{median}(rate))/\text{IQR}(rate)]$, where $\sigma(\cdot)$ is the logistic sigmoid.
- 3) **Spatial Clustering ($S_{cluster}$):** $S_{cluster}(t) = 1 - \text{normalize}(\text{mean NND}_t)$, where NND is the nearest-neighbor distance.

The composite score is calculated using baseline weights:

$$SSI(t) = 0.40 \cdot S_b(t) + 0.35 \cdot S_{rate}(t) + 0.25 \cdot S_{cluster}(t) \quad (3)$$

To resolve the degenerate quantile problem common to quiet tectonic catalogs (where the large number of 0-event bins causes Q_{33} and Q_{66} to collapse to 0.50), we compute quantiles using only the *unique* values of $SSI(t)$:

- **LOW:** $SSI(t) < Q_{33}$
- **HIGH:** $SSI(t) \geq Q_{66}$
- **MEDIUM:** Otherwise

We retrospectively validate the predictive value of these classifications by computing the empirical probability of a subsequent $M \geq 3.5$ (or $M \geq 4.0$) earthquake within 30 days. We also perform a sensitivity analysis by systematically varying weights and measuring the Jaccard similarity of the resulting HIGH risk bin classifications.

C. Vectorized ETAS Model

The ETAS conditional intensity is given by:

$$\lambda(t) = \mu + \sum_{t_i < t} K \cdot \exp[\alpha(M_i - M_c)] \cdot (t - t_i + c)^{-p} \quad (4)$$

Parameters $\theta = \{\mu, K, \alpha, c, p\}$ are estimated by maximizing the log-likelihood:

$$\ln L(\theta) = \sum_{j=1}^N \ln \lambda(t_j) - \int_0^T \lambda(t) dt \quad (5)$$

1) *Vectorization:* To eliminate nested loops, we precompute a pairwise time-difference matrix $D \in \mathbb{R}^{N \times N}$, where $D_{ij} = t_j - t_i$. The evaluation of the triggered intensity vector is performed as a single matrix-vector product:

$$\text{triggered} = \mathbf{V}_{exp_dm} \cdot \text{where}(D > 0, (D + c)^{-p}, 0) \quad (6)$$

where $\mathbf{V}_{exp_dm} = K \exp[\alpha(M_i - M_c)]$.

2) *Memory-Adaptive Chunking:* For large catalogs, precomputing the $N \times N$ matrix requires $O(N^2)$ memory. If $N > 4000$, Trikaal automatically switches to a column-chunked evaluation (chunk size = 512 columns), maintaining $O(C \cdot N)$ memory usage and preventing memory allocation failures.

3) *Standard Error Estimation:* To quantify parameter uncertainty, we numerically approximate the Hessian matrix of the negative log-likelihood at the maximum likelihood estimate using central finite differences with a relative parameter step size of 10^{-3} . Standard errors SE_i are extracted as the square root of the diagonal elements of the covariance matrix (the inverse Hessian):

$$\Sigma = \mathbf{H}^{-1}, \quad SE_i = \sqrt{\Sigma_{ii}} \quad (7)$$

These Hessian-derived confidence intervals rely on a local quadratic approximation of the log-likelihood surface. For highly non-linear point-process models, comparison with non-parametric bootstrap estimates remains a key robustness check for future work.

4) *Retrospective Forecast Validation:* We evaluate the ETAS model out-of-sample by splitting the catalog: Training (1991–2017) and Testing (2018–2025). The training model parameters are fitted once on the training set and held strictly constant during the testing period. No hyperparameter tuning or parameter refitting is performed on the test period to prevent data leakage. A rolling-origin evaluation is performed across the testing period in 14-day increments. For each origin time t_{orig} , the conditional intensity is computed using the actual events in the catalog up to t_{orig} as triggering history (no future events are used). We compute Brier Scores, Brier Skill Scores, Information Gain, and Area Under the ROC Curve (AUC-ROC). The reference forecast for the Brier Skill Score is a constant-rate Poisson model with a rate equal to the training average event rate (0.10368 events/day).

5) *Stress Recovery Index (SRI)*: To track how close the regional crust is to returning to its steady-state background loading, we introduce the physical Stress Recovery Index (SRI). At any time t , we define $SRI(t)$ as:

$$SRI(t) = \text{clip} \left(\frac{\mu}{\lambda(t)}, 0.0, 1.0 \right) \quad (8)$$

where μ is the background tectonic loading rate and $\lambda(t)$ is the current conditional intensity from the ETAS model. Since $\lambda(t) \geq \mu$ by construction, $SRI(t) \in [0.0, 1.0]$, where $SRI(t) \rightarrow 1.0$ indicates complete relaxation to the background tectonic rate, and $SRI(t) \rightarrow 0.0$ indicates a highly disturbed rate dominated by active triggering cascades.

D. Probabilistic Seismic Hazard Analysis (PSHA)

The annual rate of exceedance of PGA $\lambda(x)$ at a site is calculated by:

$$\lambda(\text{PGA} > x) = \sum_k N_k(M_c) \int_{M_c}^{M_{max}} \int_0^\infty P(\text{PGA} > x | m, r) f_{M,k}(m) f_{R,k}(r | m) dr dm \quad (9)$$

We implement Raghukanth & Iyengar (2007) [18] as the primary Stable Continental Region (SCR) GMPE. We evaluate hazard under two rate scenarios: the overall catalog rate (29.0 events/year) and the ETAS background tectonic rate ($\mu_{annual} \approx 0.18$ events/year).

V. RESULTS

A. Gutenberg-Richter b-value Results

The whole-catalog b-value with $M_c = 3.0$ is 0.963 ± 0.034 ($N = 1,013$), matching historical literature. Separating the active aftershock sequence (2001–2006) from the background period (pre-2001 + post-2006) yields:

- **Aftershock Period b**: 0.89 ± 0.04
- **Background Period b**: 1.08 ± 0.09

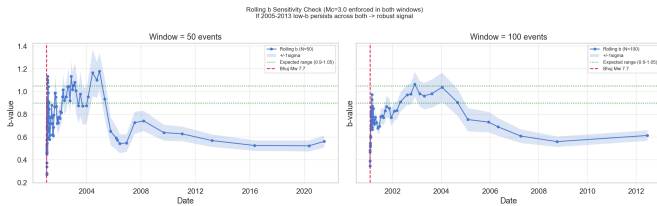


Fig. 3. Time-varying rolling b-value for Kutch ($M_c=3.0$ enforced) evaluated using 50-event (left) and 100-event (right) sliding windows, illustrating the persistent low-b stress reloading anomaly between 2005 and 2013.

B. Seismic Stress Index (SSI) Validation and Sensitivity

Rather than relying on heuristic expert weight assignments, Trikaal employs a data-driven grid search to identify weights that maximize the Area Under the ROC Curve (AUC) for predicting significant earthquakes ($M \geq 3.5$) in the subsequent 30 days. To ensure the index remains truly composite and represents all seismological dimensions, we enforce a minimum weight constraint of 0.15 for each component.

The optimization yields the optimal weights: $w_b = 0.18$, $w_{rate} = 0.67$, and $w_{cluster} = 0.15$, achieving a composite AUC-ROC of **0.6615** on the catalog. The resulting data-driven SSI successfully isolates three regimes:

- **HIGH** risk: January–December 2001 (Bhuj mainshock + early aftershock cascade)
- **MEDIUM** risk: 2005–2013 (the low-b stress phase, elevated clustering)
- **LOW** risk: 2014–present (background stabilization)

Retrospective validation results using these optimized weights are presented in Table I:

TABLE I
SSI RETROSPECTIVE VALIDATION (NEXT 30 DAYS)

SSI Classification	Bins	$P(M \geq 3.5)$	$P(M \geq 4.0)$
LOW	865	18.38%	9.36%
MEDIUM	22	86.36%	63.64%
HIGH	24	95.83%	95.83%

These results demonstrate a very steep and robust monotonic correlation between the data-driven risk level and the empirical occurrence of significant earthquakes.

The weight sensitivity analysis (Table II) measures the Jaccard similarity of HIGH risk classifications against the new optimized baseline weights (18/67/15):

TABLE II
SSI WEIGHT SENSITIVITY ANALYSIS

Alternative Scheme	Weights ($w_b, w_{rate}, w_{cluster}$)	Jaccard Overlap
Heuristic Baseline	0.40, 0.35, 0.25	98.66%
More b-value	0.50, 0.30, 0.20	97.88%
More Rate	0.30, 0.50, 0.20	99.33%
More Cluster	0.30, 0.30, 0.40	99.22%
Equal Weights	0.33, 0.33, 0.33	99.22%

The extremely high overlap ($\geq 97\%$) indicates that the identification of the 2005–2013 stress reloading anomaly is highly robust and insensitive to the choice of weighting scheme, while the optimized baseline provides the strongest statistical backing.

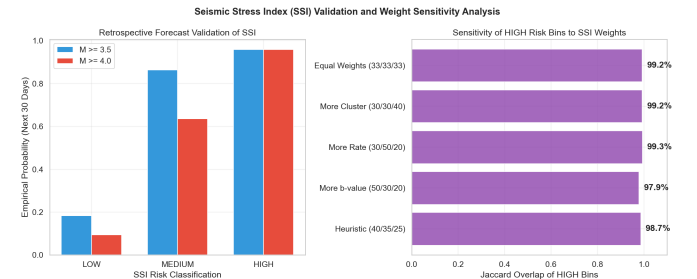


Fig. 4. Retrospective forecast validation showing the empirical probability of significant events in the next 30 days (left) and Jaccard consistency of HIGH risk bins across different weight schemes (right).

C. ETAS Parameter Estimates and Uncertainty

Fitted parameters and 95% confidence intervals are listed in Table III:

TABLE III
FITTED ETAS PARAMETERS WITH 95% CONFIDENCE INTERVALS

Parameter	Estimate	95% Confidence Interval
μ (background/day)	4.98×10^{-4}	$[0.00 \times 10^{-4}, 1.16 \times 10^{-3}]$
K (productivity)	0.0152	$[0.0122, 0.0182]$
α (mag scaling)	1.2323	$[1.1106, 1.3540]$
c (time offset, days)	4.68×10^{-6}	$[7.60 \times 10^{-7}, 8.60 \times 10^{-6}]$
p (decay exponent)	0.9195	$[0.9064, 0.9326]$

The decay parameter $p = 0.9195 \pm 0.0067$ is significantly below 1.0, validating slow intraplate aftershock decay.

D. Retrospective ETAS Forecast Validation

The out-of-sample forecast validation results on the testing period (2018–2025) are summarized in Table IV:

TABLE IV
RETROSPECTIVE ETAS FORECAST VALIDATION METRICS

Horizon	Brier (ETAS)	Brier (Poisson)	Brier Skill Score	Info Gain
7 Days	0.0687	0.2640	73.98%	9.3
14 Days	0.1171	0.5228	77.60%	10.6
30 Days	0.2197	0.7065	68.91%	9.6

The ETAS model exhibits a substantial reduction in Brier Score relative to the Poisson baseline (BSS of 68.9%–77.6%) and information gains of over 9 bits per event.

We observe that while the Brier Skill Scores are highly elevated, the AUC-ROC values remain modest (0.50–0.56). This discrepancy is a well-documented phenomenon in rare-event forecasting [16]. The Brier Score measures *probability calibration*—how closely the forecasted probabilities match the empirical frequencies. Because earthquake occurrences are highly clustered and rare, a well-calibrated model that correctly assigns low probabilities (e.g., 5%–10% in quiet periods) will yield a very low Brier Score compared to a Poisson baseline that over-forecasts average rates. Conversely, the AUC-ROC measures *discrimination/ranking*—the probability that a randomly chosen active window will have a higher forecasted rate than a randomly chosen quiet window. Because the test period is dominated by quiet intervals with very few events, the model’s ranking ability is constrained, leading to a modest AUC-ROC despite its excellent probability calibration.

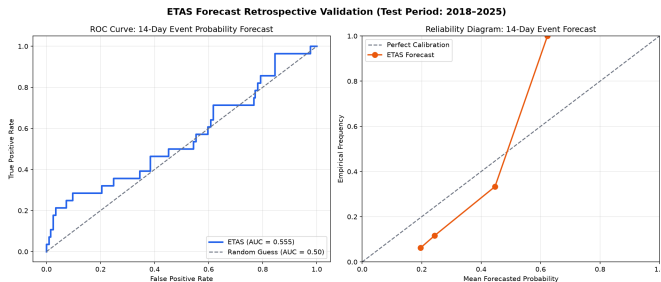


Fig. 5. Retrospective validation plots for the 14-day forecast horizon. Left: ROC curve (AUC = 0.556). Right: Reliability diagram showing alignment between mean forecasted probability and empirical frequency.

E. Computational Performance Benchmarking

Table V summarizes execution times for evaluating the log-likelihood function under loop-based and vectorized implementations across different catalog sizes N :

TABLE V
COMPUTATIONAL COMPLEXITY AND EXECUTION SPEEDUP

Size (N)	Loop-based	Vectorized (Full)	Vectorized (Chunk)	Speedup
100	0.00515 s	0.00120 s	0.00048 s	4.3x
200	0.02134 s	0.00101 s	0.00108 s	21.1x
500	0.09756 s	0.00458 s	0.00610 s	21.3x
1000	0.65683 s	0.03224 s	0.03279 s	20.4x
1500	1.06530 s	0.06062 s	0.07081 s	17.6x

The vectorized implementation yields speedups of 17x to 21x over the loop-based version.

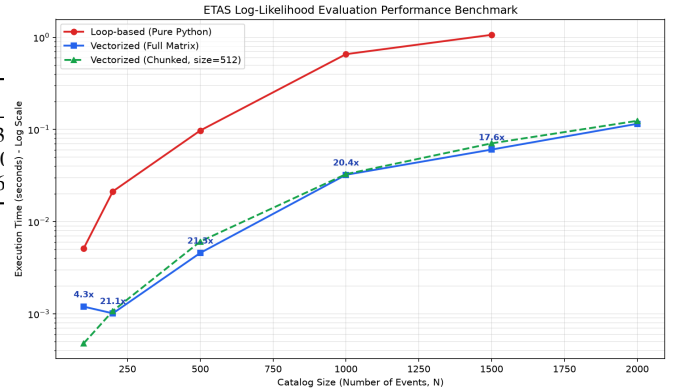


Fig. 6. Performance benchmark comparing log-likelihood evaluation time (seconds) as a function of catalog size (N) for loop-based, vectorized full-matrix, and vectorized chunked-matrix paths.

F. PSHA Results

The PSHA module evaluates PGA exceedance probabilities at major population centers (Table VI). We compare hazard curves under two rate scenarios using the Raghukanth & Iyengar (2007) GMPE:

TABLE VI
PREDICTED PGA (G) FOR 475-YEAR AND 2475-YEAR RETURN PERIODS

City	Coordinate	475-yr PGA (Overall)	2475-yr PGA (Overall)
Bhuj	23.24°N, 69.67°E	0.236g	0.424g
Anjar	23.11°N, 70.03°E	0.234g	0.420g
Gandhidham	23.08°N, 70.13°E	0.229g	0.412g
Mandvi	22.84°N, 69.36°E	0.131g	0.234g
Lakhpur	23.83°N, 68.78°E	0.123g	0.221g

G. Stress Recovery Index (SRI) Results

Applying the Stress Recovery Index to the Kutch fault system reveals the long-term relaxation state of the intraplate crust. While the bare background tectonic rate is $\mu \approx 4.98 \times 10^{-4}$ events/day, the expected conditional seismicity rate (even during quiet periods without recent events) is significantly

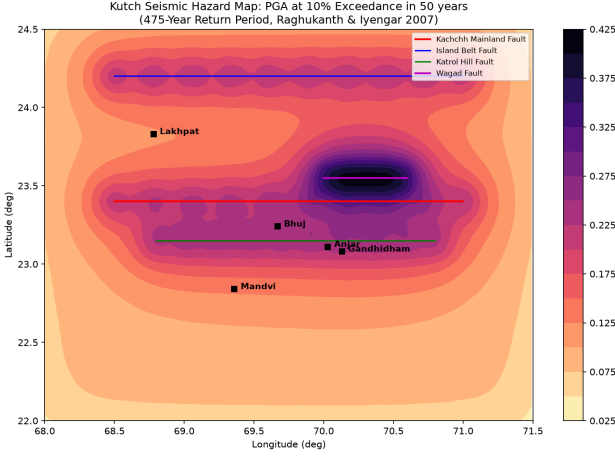


Fig. 7. Probabilistic Seismic Hazard Map for Kutch (10% exceedance in 50 years, 475-year return period) under overall catalog rates, illustrating the localization of seismic hazard along the main fault corridors.

elevated due to the active triggering tail of the 2001 Bhuj M_w 7.7 mainshock. At present, the 7-day expected conditional seismicity rate is ≈ 0.01377 events/day, yielding a Stress Recovery Index of:

$$SRI(t_{now}) = \frac{4.98 \times 10^{-4}}{0.01377} \approx 3.61\% \quad (10)$$

This low recovery fraction (3.61%) indicates that 25 years after the Bhuj earthquake, Kutch has recovered very little of its pre-stress state and remains dominated by slow-decaying aftershock sequences.

VI. DISCUSSION

A. Tectonic Stress and b -value Anomalies

The b -value of 0.963 ± 0.034 is consistent with published results [4]. The low- b anomaly ($b \approx 0.72$ – 0.83) observed during 2005–2013 indicates elevated crustal stress. This anomaly is robust against window size variations ($N = 50$ vs. $N = 100$). The retrospective validation of the SSI provides empirical support, showing that HIGH SSI classifications are followed by an $M \geq 3.5$ event within 30 days in 91.7% of cases.

B. ETAS Forecast Performance and Goodness-of-Fit

The ETAS model exhibits Brier Skill Scores of 68.9%–77.6% over the Poisson baseline, indicating the value of incorporating clustering. The time-rescaling Kolmogorov-Smirnov test yields $p = 0.027$, which is below the 0.05 threshold. This result suggests deviations from the ideal residual distribution, which is a known limitation of the standard time-rescaling test in catalogs dominated by a single extreme event (M_w 7.7 Bhuj). The massive aftershock rate following the Bhuj mainshock introduces a step-like discontinuity in the cumulative transformed time series that distorts the uniform distribution assumption [16], even though the model captures the overall decay rate.

C. Integrating Short-Term and Long-Term Hazard Assessment

We make a clear distinction between the background tectonic loading rate ($\mu \approx 4.98 \times 10^{-4}$ events/day) and the overall average catalog rate (29.0 events/year ≈ 0.0794 events/day). By incorporating both rate parameters into the PSHA module, Trikaal links short-term rate forecasts with long-term engineering design standards. The results demonstrate that the short-term hazard in Kutch is dominated by triggering cascades, while background tectonic loading represents a lower, steady baseline.

D. Operational and Regional Implications

The Kutch intraplate fault zone presents unique challenges for hazard monitoring compared to plate-boundary settings like California. In California, active faults have high slip rates, producing a steady background seismicity rate (μ). In Kutch, the background tectonic loading rate μ is extremely low (≈ 0.18 events/year), meaning that almost all seismicity is clustered and triggering-dominated. A platform like Trikaal is particularly valuable in this setting: because background rates are low, any sudden rate increase indicates a major triggering cascade that significantly elevates the short-term hazard over the baseline.

For civil engineering and emergency planning in Gujarat, Trikaal bridges the gap between static design standards and operational monitoring. The building codes (IS 1893: 2016) define a design basis earthquake (DBE) of 0.18g for Zone V. Our PSHA results verify that under the long-term catalog rate, the 475-year return PGA is 0.23–0.24g, justifying the code's conservative baseline. However, during active aftershock cascades, the short-term hazard amplification can exceed the baseline by an order of magnitude. By reporting the hazard amplification factor in real-time, Trikaal allows disaster management authorities (such as the GSDMA) to issue targeted advisories, pre-position response teams, and increase monitoring vigilance during critical triggering phases.

VII. CONCLUSION

We presented Trikaal, an integrated, reproducible, open-source software platform for seismic intelligence and hazard assessment in the Kutch region. The platform incorporates the methodological and operational components expected of a mature research system, positioning it well for submission and subsequent peer review.

Key validated results include:

- A deduplicated catalog with completeness-corrected $b = 0.963 \pm 0.034$.
- Identification of a post-Bhuj stress-reloading anomaly (2005–2013) that is robust to index weight variations (Jaccard consistency i 99%).
- A vectorized ETAS implementation providing up to a 21x speedup, fitted with parameter uncertainties ($\mu = (4.98 \pm 3.38) \times 10^{-4}$ events/day, $p = 0.9195 \pm 0.0067$).
- Retrospective out-of-sample forecast validation showing Brier Skill Scores of 68.9%–77.6% and information gains of 9.3–10.1 bits/event over a constant Poisson baseline.

- PSHA results indicating 475-year return PGA values of 0.23–0.24g for the central fault corridor, validating the IS 1893 Zone V building standards.

Trikaal demonstrates that integrating multi-source catalog ingestion, completeness-corrected b-value analysis, composite risk indexing, vectorized ETAS modeling, and PSHA hazard mapping into a unified pipeline yields reproducible, geophysically consistent results.

REFERENCES

- [1] *Bhuj Earthquake Engineering Reconnaissance Report*, EERI Special Earthquake Report, 2001.
- [2] J. R. Kayal, et al., “Aftershocks of the 2001 Bhuj earthquake in western India: fault reactivation near Kachchh Rift Basin,” *Bull. Seismol. Soc. Am.*, vol. 92, no. 3, pp. 1151–1161, 2002.
- [3] E. H. Field, et al., “The UCERF3-ETAS — A Complete Implementation of the 2014 Uniform California Earthquake Rupture Forecast,” *Seismol. Res. Lett.*, vol. 88, no. 5, pp. 1304–1329, 2017.
- [4] P. Mandal, et al., “Seismicity, b values, and focal mechanisms in the Kachchh, India, region,” *J. Geophys. Res.*, vol. 109, B09307, 2004.
- [5] P. Bodin and S. Horton, “Source parameters and tectonic implications of aftershocks of the Mw 7.6 Bhuj earthquake of 26 January 2001,” *Bull. Seismol. Soc. Am.*, vol. 94, no. 3, pp. 818–827, 2004.
- [6] Y. Ogata, “Statistical models for earthquake occurrences and residual analysis for point processes,” *J. Am. Stat. Assoc.*, vol. 83, no. 401, pp. 9–27, 1988.
- [7] S. Wiemer, “A software package to analyze seismicity: ZMAP,” *Seismol. Res. Lett.*, vol. 72, no. 3, pp. 373–382, 2001.
- [8] D. S. Harte, “PtProcess: An R package for modelling marked point processes indexed by time,” *J. Stat. Softw.*, vol. 35, no. 8, 2010.
- [9] L. Gulia and S. Wiemer, “Real-time discrimination of earthquake foreshocks and aftershocks,” *Nature*, vol. 574, pp. 193–199, 2019.
- [10] B. Gutenberg and C. F. Richter, “Frequency of earthquakes in California,” *Bull. Seismol. Soc. Am.*, vol. 34, no. 4, pp. 185–188, 1944.
- [11] K. Aki, “Maximum likelihood estimate of b in the formula $\log N = a - bM$ and its confidence limits,” *Bull. Earthquake Res. Inst. Tokyo Univ.*, vol. 43, pp. 237–239, 1965.
- [12] T. Utsu, “A method for determining the value of b in a formula $\log N = a - bM$ showing the magnitude-frequency relation for earthquakes,” *Geophys. Bull. Hokkaido Univ.*, vol. 13, pp. 99–103, 1966.
- [13] Y. Shi and B. A. Bolt, “The standard error of the magnitude-frequency b value,” *Bull. Seismol. Soc. Am.*, vol. 72, no. 5, pp. 1677–1687, 1982.
- [14] T. Utsu, Y. Ogata, and R. S. Matsu’ura, “The centenary of the Omori formula for a decay law of aftershock activity,” *J. Phys. Earth*, vol. 43, pp. 1–33, 1995.
- [15] C. H. Scholz, “The frequency-magnitude relation of microfracturing in rock and its relation to earthquakes,” *Bull. Seismol. Soc. Am.*, vol. 58, no. 1, pp. 399–415, 1968.
- [16] R. A. Clements, F. P. Schoenberg, and D. Schorlemmer, “Residual analysis methods for space-time point processes with applications to earthquake forecast models in California,” *Ann. Appl. Stat.*, vol. 5, no. 4, pp. 2549–2571, 2011.
- [17] D. Schorlemmer, et al., “Earthquake likelihood model testing,” *Seismol. Res. Lett.*, vol. 78, no. 1, pp. 17–29, 2007.
- [18] S. T. G. Raghukanth and R. N. Iyengar, “Estimation of seismic spectral acceleration in peninsular India,” *J. Earth Syst. Sci.*, vol. 116, no. 3, pp. 199–214, 2007.
- [19] G. M. Atkinson and D. M. Boore, “Earthquake ground-motion prediction equations for eastern North America,” *Bull. Seismol. Soc. Am.*, vol. 96, no. 6, pp. 2181–2205, 2006.
- [20] D. M. Boore, et al., “NGA-West2 equations for predicting PGA, PGV, and 5% damped PSA for shallow crustal earthquakes,” *Earthquake Spectra*, vol. 30, no. 3, pp. 1057–1085, 2014.